Yong Guk Kang

Research Professor, Imaging Intelligence Laboratory, School of Mechanical Engineering, Seoul National University

Optical systems and computational imaging researcher focused on physics-informed simulation, inverse optical design, lensless imaging, and machine-learning assisted image reconstruction. I translate research problems into imaging system models, reconstruction pipelines, and experimental strategies for optical and biomedical applications.

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RESEARCH FOCUS

Computational imaging and inverse optical design; Lensless and phase-coded imaging systems; AI-assisted image reconstruction and virtual staining; Quantitative phase, interferometric, and coherent imaging; Physics-informed modeling of optical imaging systems

SELECTED CAPABILITIES

Computational imaging: Wave-optics simulation, Inverse problems and image reconstruction, Machine learning assisted optical design and reconstruction

Optical imaging: Quantitative phase imaging, Interferometric microscopy, Optical coherence microscopy, Nonlinear microscopy

Tools: MATLAB, Python, PyTorch, ZEMAX, Fusion 360, ImageJ

TECHNICAL SCOPE

Computational imaging: Wave optics simulation, Lensless imaging, Inverse design, Optical image reconstruction, Physics-based noise modeling, Interferometric imaging, Image deconvolution, Virtual staining

Optical imaging modalities: Quantitative phase imaging, Interferometric microscopy, Nonlinear microscopy, Optical coherence tomography, Optical coherence microscopy, Off-axis holographic microscopy

Computational tools: MATLAB, Python, PyTorch, ZEMAX, ImageJ

Experimental implementation: Optical system prototyping, Microfabrication workflows, 3D CAD and rapid prototyping

EXPERIENCE

Research Professor, Seoul National University, School of Mechanical Engineering (2025-04-present)

- Developing computational imaging frameworks that combine optical system modeling, inverse design, and data-driven reconstruction.
- Investigating lensless and phase-coded imaging strategies for high-dynamic-range and information-rich optical measurements.
- Applying physics-informed simulation and machine learning to jointly reason about optical encoding and image reconstruction.

Researcher, Yonsei University, School of Electrical and Electronic Engineering (2024-04-2025-02)

- Contributed to computational imaging and lensless imaging research in the Imaging Intelligence Laboratory.
- Worked on phase-coded optical systems, wave-optics modeling, and reconstruction-oriented imaging methods.
- Collaborated on compact optical imaging platforms and biomedical imaging applications.

Postdoctoral Researcher, Korea University, School of Biomedical Engineering (2022-09-2024-03)

- Advanced reflection phase microscopy for efficient 3D quantitative imaging with reduced axial scanning.
- Developed computational refocusing and field synthesis strategies for reflection phase measurements.
- Integrated optical modeling and experimental validation for interferometric biomedical imaging.

Research Lecturer, Korea University, School of Medicine (2020-10-2022-08)

- Developed optical coherence microscopy approaches for quantifying cell-ECM interactions and intracellular dynamics.

- Applied temporal-dynamics contrast imaging to live-cell and tissue-scale biomedical measurements.
- Connected optical imaging measurements with quantitative analysis of mechanical interactions in biological matrices.

EDUCATION

Ph.D. in Biomedical Engineering, Korea University

B.S. in Biomedical Engineering, Korea University

PROJECTS

Lensless and HDR computational imaging. Computational imaging research on lensless and phase-coded acquisition, with emphasis on optical encoding, inverse design, and robust reconstruction under challenging measurement conditions.

Reflection phase microscopy. Quantitative reflection phase microscopy for efficient 3D surface and biomedical imaging, combining interferometric measurement design with computational refocusing and field synthesis.

Optical coherence microscopy for cell-ECM dynamics. Optical coherence microscopy and temporal-dynamics contrast imaging for studying cell, tissue, and extracellular matrix interactions.

Live-cell nonlinear microscopy for matrix deformation analysis. Nonlinear microscopy and quantitative image analysis for studying collagen matrix deformation and cell-ECM mechanical interactions.

SELECTED JOURNAL ARTICLES

Geometry-aware phase compensation for sampling-efficient angular spectrum method, Optics Express (2026)

Wide-area topography using reflection phase microscopy for surface inspection, Optics Express (2025)

Semi-supervised virtual staining using learned-illumination Fourier ptychography for high-speed label-free histopathology, Journal of Physics: Photonics (2025)

Cardiac fibroblast-mediated ECM remodeling regulates maturation in an in vitro 3D engineered cardiac tissue, Journal of Tissue Engineering (2025)

Laser-tissue interaction simulation considering skin-specific data to predict photothermal damage lesions during laser irradiation, Journal of Computational Design and Engineering (2023)

Three-dimensional imaging in reflection phase microscopy with minimal axial scanning, Optics Express (2023)

Optical coherence microscopy with a split-spectrum image reconstruction method for temporal-dynamics contrast-based imaging of intracellular motility, Biomedical Optics Express (2023)

Development of a 3-D Physical Dynamics Monitoring System Using OCM with DVC for Quantification of Sprouting Endothelial Cells Interacting with a Collagen Matrix, Materials (2020)

Reflection Phase Microscopy by Successive Accumulation of Interferograms, ACS Photonics (2019)

Quantification of focal adhesion dynamics of cell movement based on cell-induced collagen matrix deformation using second-harmonic generation microscopy, Journal of Biomedical Optics (2018)

Single-shot digital holographic microscopy for quantifying a spatially-resolved Jones matrix of biological specimens, Optics Express (2016)

SELECTED CONFERENCE PAPERS

High-speed virtual re-staining via near-infrared quantitative phase imaging, Proceedings of SPIE Advanced Biophotonics Conference (2026)

A compact, bite-down lensless imager for 3D fluorescence mapping of occlusal dental plaque, Proceedings of SPIE (2026)

Non-Paraxial Layered Diffraction Simulator for Inverse Design of Lensless Imaging Optics, Optica Imaging Congress 2025 (2025)

PATENTS AND RECOGNITION

Funding: Creative and Challenging Research Base Support (2021R1I1A1A01048370)

Fellowship: Global Ph.D. Fellowship (2015H1A2A1030048)

Patent: Head mount system for providing surgery support image (US 11,356,651)

Patent: Reflection phase microscopy system for 3D image acquisition (10-2024-0081557)

Patent: Phase mask fabrication method based on grayscale lithography (10-2024-0161144)